Naval Information Warfare Center, Pacific
Atmospheric Propagation Branch, Code 55280
rfppas@niwc.navy.mil

ATAK RF Propagation Plugin v5.0+
User Manual

Table of Contents

1 Plug-in Overview 2
1.1 Plugin Description 2
1.2 When to use ATAK Plugin 2
2 Installing the Plug-In & ATAK 3
3 Modes of Operation 3
3.1 Coverage 3
3.2 Overlays 4
3.3 Explore Loss 5
3.4 Point-to-Point (P2P) 6
3.5 Route 7
4 RF Parameters 8
5 Meteorology 8
5.1 What is refractivity? 9
5.2 Why add refractivity to the model? 9
5.3 When to avoid refractive effects 10
6 Exporting refractivity from Builder 11
6.1 **Import Weather Data** 11
6.2 Select the time range 12
6.3 Create a refractivity plot 13
6.4 Enable automatic plot exporting 14
6.5 Enable plotting and play the scenario to export all forecast times 15
6.6 Push to ATAK device or TAK Server 16
7 Importing METOC Refractivity data in the plugin 16

1 Plug-in Overview

The ATAK RF Propagation plugin runs a simulation of radio frequency (RF) propagation, accounting for terrain and weather effects. It calculates "propagation loss" which, when combined with knowledge of transmitter and receiver gains, enables a link budget estimate. The RF model used in this plugin is the Advanced Propagation Model, developed and maintained by NIWC Pacific, Atmospheric Propagation Branch 55280: rfppas@niwc.navy.mil.

A strong RF link budget informs tactical decisions such as waveform selection, installation locations, or when and where to expect degraded connectivity.

USE CASE EXAMPLE

The MPU5 L-Band radio has a transmit power of 35 dBm (3.3W) and a receiver sensitivity of -95dBm, operating around 2.3GHz. A link can persist anywhere the model predicts propagation loss is below 130 dB.

This manual goes through the installation and use of the plugin. Then an overview of refractivity profiles is provided followed by the process of downloading numerical weather prediction (NWP) data in NRL Builder using a laptop. Finally, exporting refractivity profiles and transferring those files to an Android device for use in the ATAK RF Propagation plugin is explained.

1.1 Plugin Description

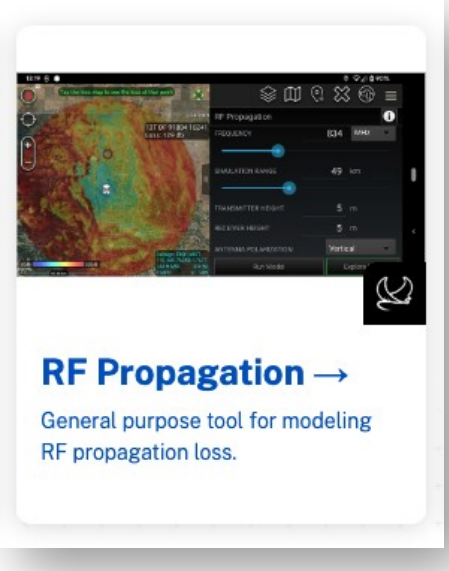
General purpose tool for modeling RF propagation loss. Plan EMW effects or link quality along routes, understand radar coverage, optimize sensor emplacements for SIGINT missions, design long-haul COMMS links, determine EW vulnerabilities or ES opportunities. If available, incorporate atmospheric refractivity effects to completely characterize an operating environment. Valid 3 MHz - 18 GHz. Uses DTED terrain data on device. Does not account for HF skywave (groundwave only). The plugin can be run with or without METOC inputs.

1.2 When to use ATAK Plugin

Complex electromagnetic scenarios can be modeled using the Naval Research Lab's Scenario Builder using a laptop. It includes advanced features like HF skywave and radio network modeling. The ATAK RF Propagation Plugin is a very simple tool for "back of the envelope" calculations when the full modelling capabilities may not be necessary or possible. This is especially useful for expeditionary use cases when lightweight and dismounted operations are needed.

All memory from the plugin will be reset when the ATAK app closes. Save any screenshots or information you need prior to leaving ATAK.

2 Installing the Plug-In & ATAK



To install this plugin, you will need an Android device and an account from https://tak.gov. The plug-in and the ATAK OS will be downloaded separately.

After creating a TAK account, navigate to **Products → All Plugins.** Select & download the "RF Propagation" plugin.

The plug-in is compatible with ATAK 5.x MIL Flavor only.

Select **Products → ATAK-MIL.** Select ATAK-MIL 5.1.0 from the drop down menu. Download the Orange CORE CIV and the Blue MIL Flavor.

Load the packages onto the Android device to install.

Example Filenames:
PlugIn: ATAK-Plugin-rf-propagation-5.1.0.1.beta2024-4866053f-5.1.0-mil-release.apk
CIV: release ATAK-5.1.0.24-cd7e8bb5-civ-release.apk
MIL: ATAK-SystemPlugin-FlavorPlugin-5.1.0.24-cd7e8bb5-5.1.0-mil-release.apk

A NOTE ON TAK CERTS

TAK certificates are not needed to install ATAK or the RF Propagation plugin. TAK certs allow ATAK users to join a network and connect with others.

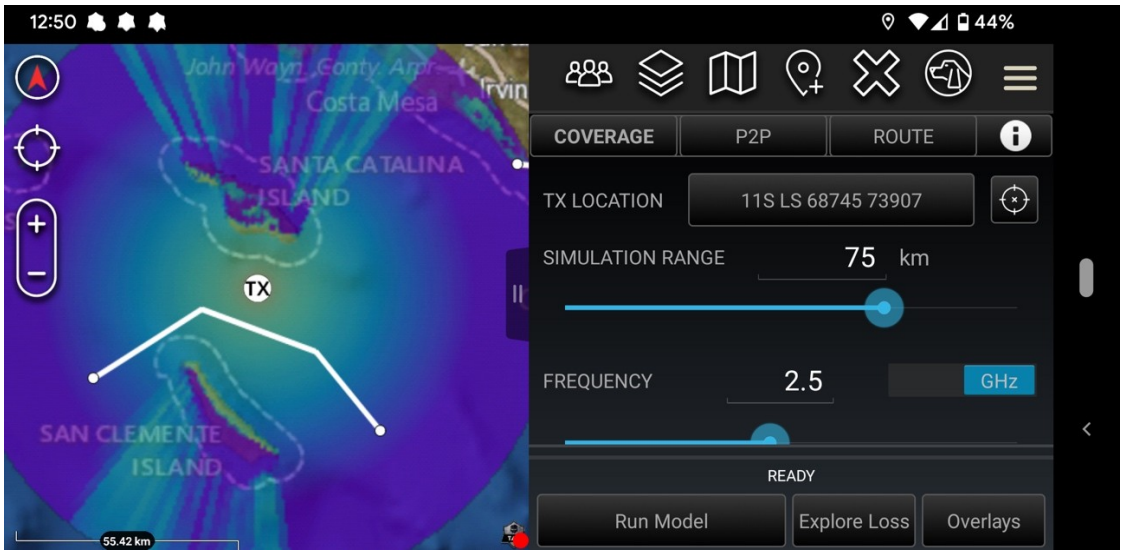
3 Modes of Operation

There are 3 modes of operation for this plugin. These options are always shown at the top of the display. The currently selected mode is in **bold.**

The information button will show a quick reference guide, as well as contact information of the NIWC PAC development team.

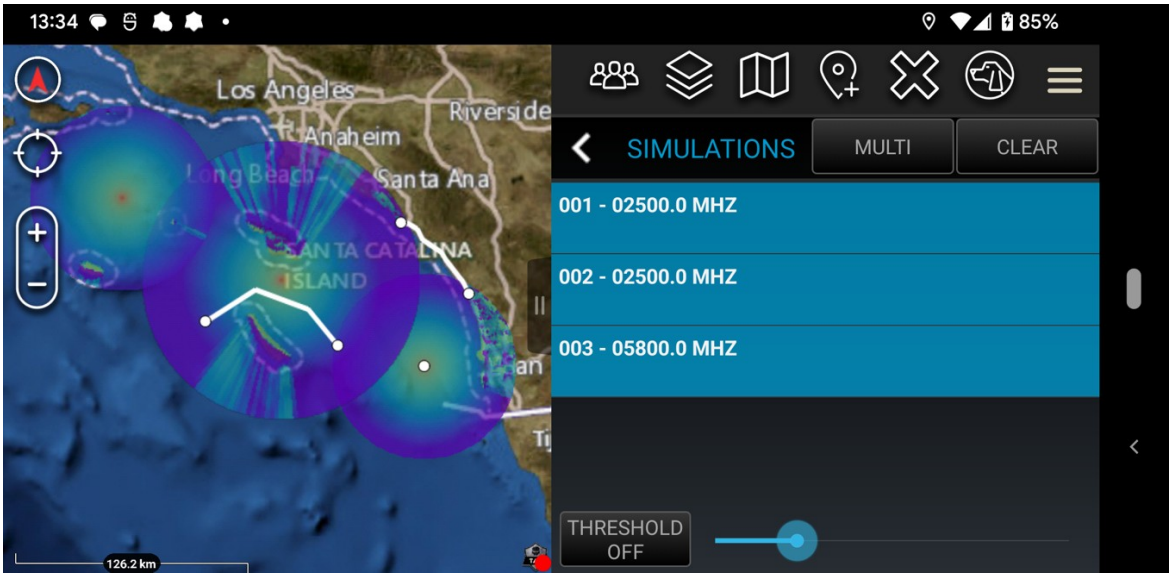
3.1 Coverage

Simulates radio propagation from a transmitter in all directions. The resolution is currently fixed at 2-degree azimuth steps. On completion, the simulation results are rendered on the map, and a new item is added to the Overlays view.

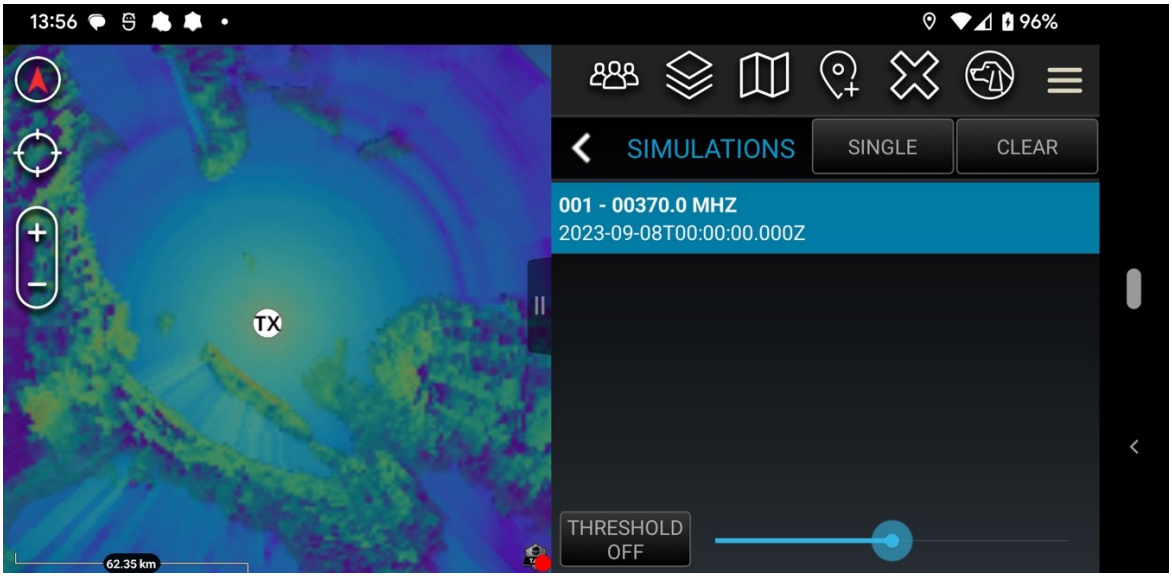


3.2 Overlays

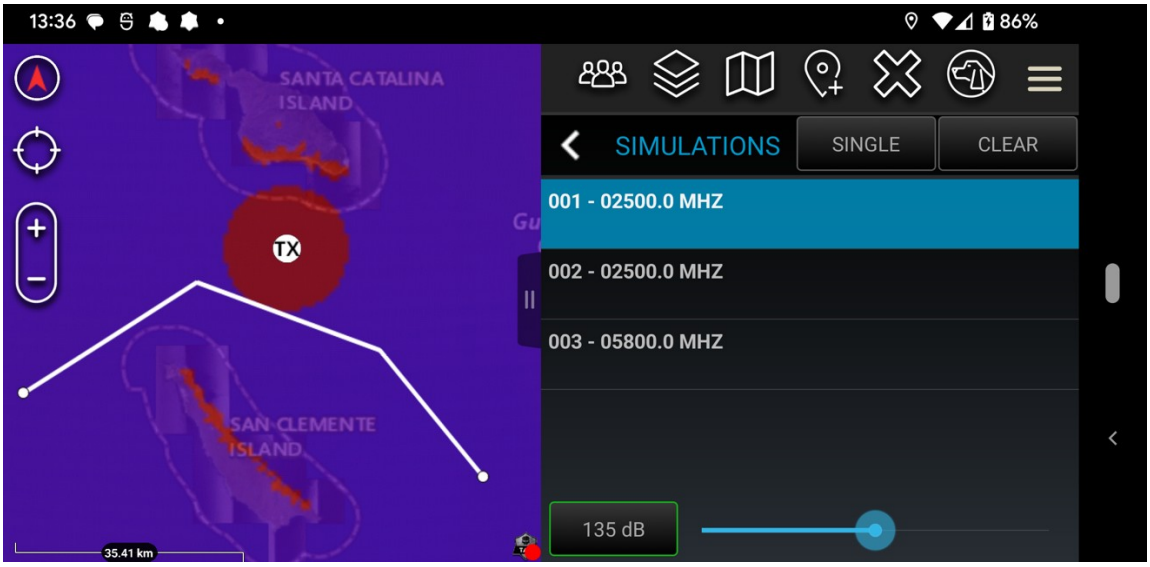
Displays a list of previous simulations available for display. The CLEAR button will delete all simulations from memory. When in MULTI mode, you can display multiple overlays at once. When in SINGLE mode, you can quickly switch between individual overlays. SINGLE mode can be useful for quick A/B comparisons.



If a METOC file is loaded, the Overlays view will display the model forecast date/time.

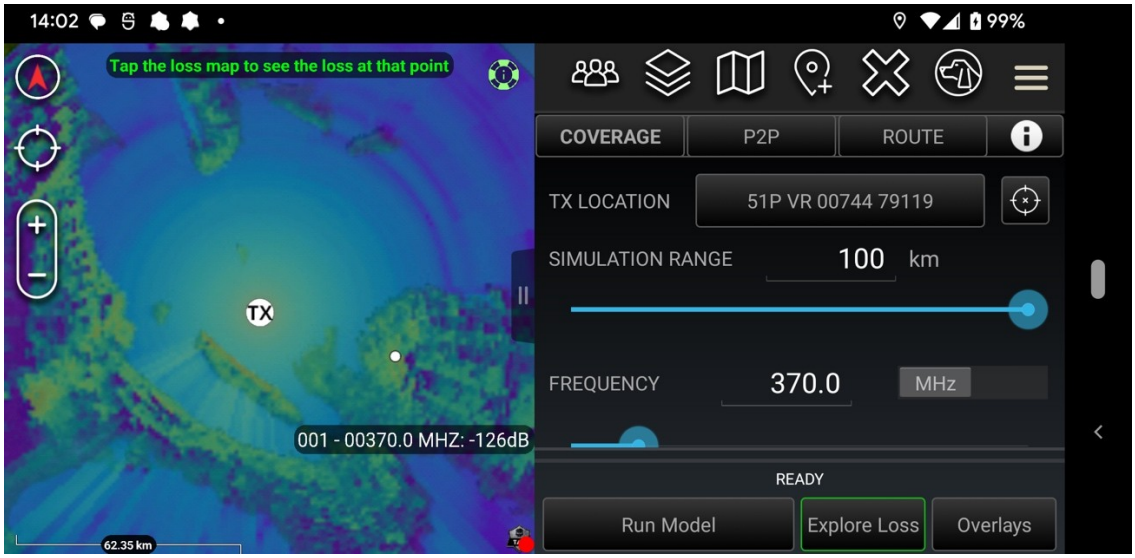


The THRESHOLD feature will re-render all propagation overlays to a binary colormap, according to the value set with the slider. For example, if you know that your link budget threshold is –135dB, then the plot will show the regions above this critical value.



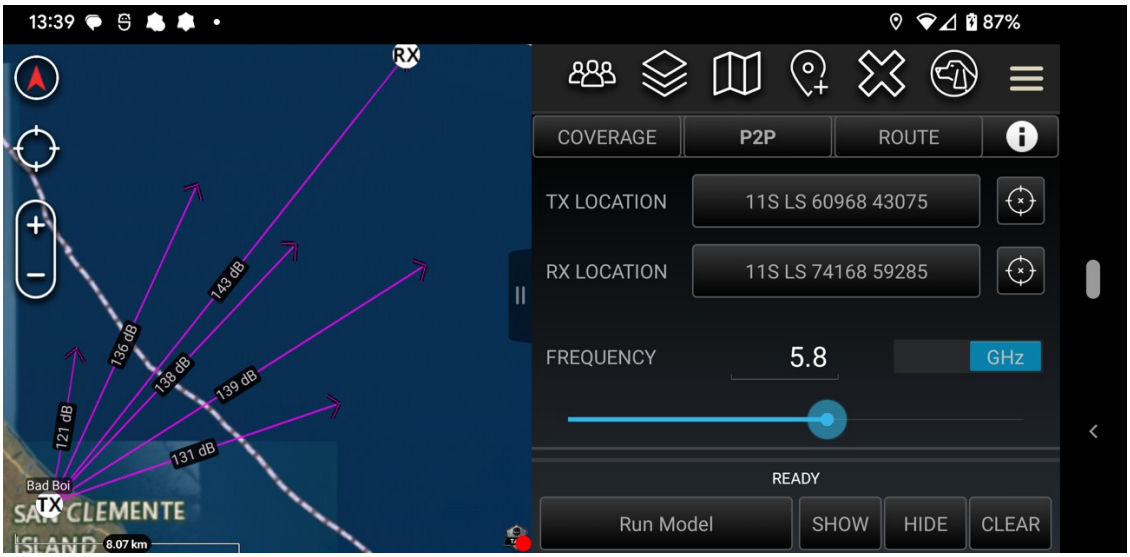
3.3 Explore Loss

This tool enables an overlay touch-query. If you touch somewhere on a coverage plot, it will display the loss at that point. If multiple overlays cover that point, then the losses from each transmitter will display at the same time.



3.4 Point-to-Point (P2P)

If you know the exact location of the transmitter and receiver, you can run a quick simulation to get the loss between those points. Each result will be displayed as an arrow on the map connecting transmitter to receiver.



3.4.1 Show/Hide/Clear

These buttons show, hide, and clear all the arrows from P2P simulations. Clear will delete all arrows from memory.

3.5 Route

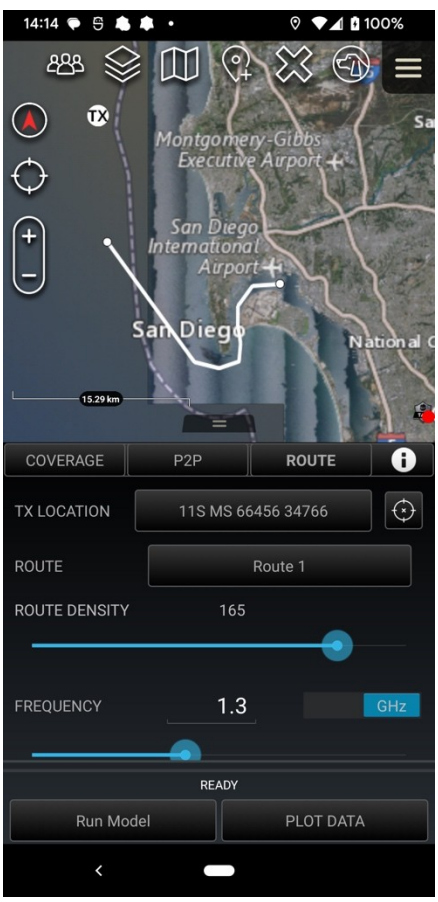
You can model a radio link from a fixed point to a receiver moving along a route. The sampling density can be set with the slider. A higher density can help you visualize propagation over complex terrain, but will make the simulation take longer. This can be used to predict coverage or dead zones for a mobile link.

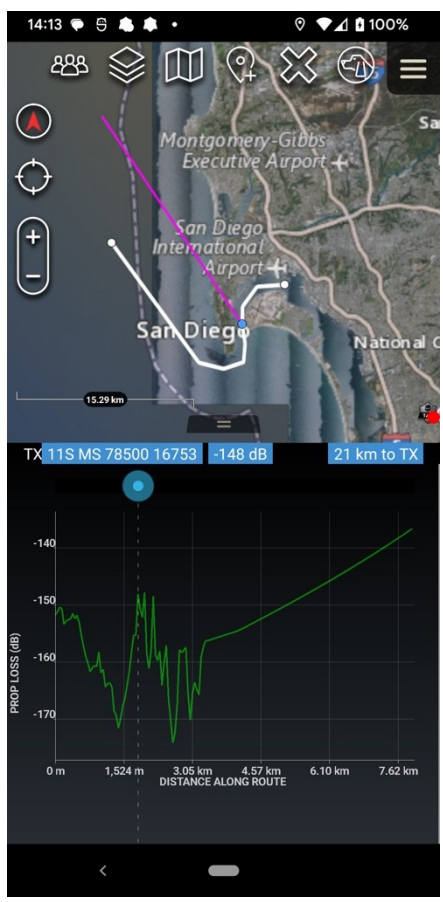
3.5.1 Choose Route

This tool queries the core ATAK route list. If you don't have any routes, this menu will be blank. You need to exit the RF Propagation Plugin and use the Routes tool to modify or add routes. You may need to restart ATAK if you modify or delete a route after simulating the RF propagation along that route.

3.5.2 Plot Data

The route simulation is displayed on a graph. The x-axis shows the progress along the route, from start to finish. The y-axis shows the propagation loss at that point. You can move the blue dot along the top of the graph to draw a line from the transmitter to that point on the route. The values highlighted in blue at the top of the chart: the location on the route, the loss at that point, and the distance to the transmitter.





4 RF Parameters

Frequency: this slider automatically updates the MHz/GHz switch. You can also type the frequency manually.

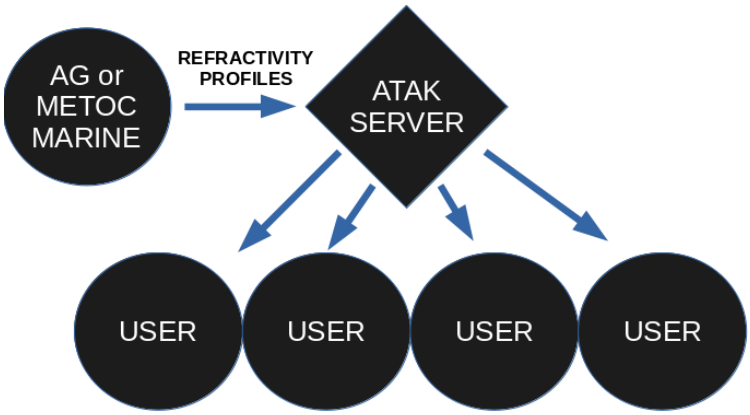
Antenna heights are above ground level (AGL), according to DTED. If DTED is missing, or the antenna is over the ocean, these values are above mean sea level (MSL). If you are operating over terrain but are missing DTED terrain data, answers will be wildly inaccurate.

Polarization mainly affects reflections and terrain diffraction. If you aren't sure, leave this parameter alone – impact is relatively low.

Note: Antennas are assumed isotropic; 0 dBi in all directions. If you know the gain of your antenna, add it to your link budget and set thresholds accordingly.

5 Meteorology

Atmospheric refractivity changes by the hour, and can heavily impact the way electromagnetic energy travels through the air. If you get the correct type of weather prediction files, you can use them in ATAK for radio propagation modeling.



WORKFLOW

A METOC specialist downloads meteorology data, converts/uploads refractivity data to a TAK server, improving modeling capabilities of ATAK users running the RF Propagation Plugin. Delete past METOC data files on the TAK Server to avoid filling up the drive space.

5.1 What is refractivity?

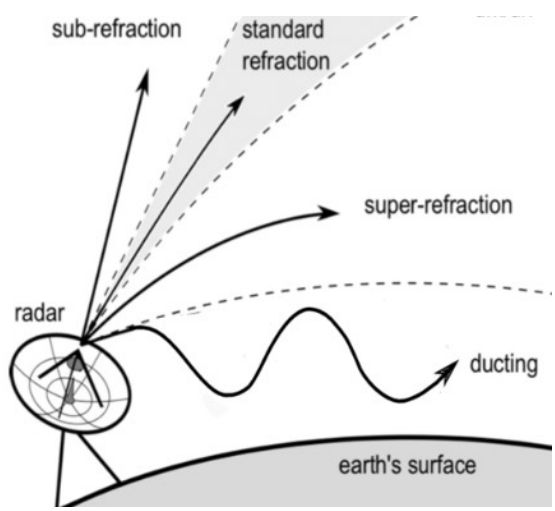
The refractive index is a number that describes the speed of light in a material. While it never goes faster than the maximum speed of light in a vacuum, it slows down when passing through matter (dirt, water, gasses, food, whatever). In this case, “light” means any electromagnetic energy, like radios, lasers, ground-penetrating radars, microwaves, etc.

Whenever there is a change in the speed of a wavefront, it changes direction. Just like your car gets pulled to one side when hitting a deep puddle on the side of the road, a higher refractive index slows down electromagnetic energy causing it to bend towards the slower path. Look up “Snell’s Law” for more info.

5.2 Why add refractivity to the model?

Due to certain weather patterns, the atmosphere in your AOR can have refractive index changes that can affect radio and radar systems. These changes cause electromagnetic energy to bend either upward or downward.

- If it bends upward, the atmosphere is referred to as “subrefractive”
  - Communication links drop more often
  - Radios have worse range
  - Radars pick up less surface targets
- If energy is bent downward, you have superrefractive or “ducting” conditions
  - Communication links get extended range
  - Pick up distant radio signals you don’t usually receive
  - Radar surface targets show up at extended range
  - Radars may display “clutter rings” as the energy skips along the surface



5.3 When to ignore refractive effects

If you're planning a mission with any of the following characteristics, atmospheric refractivity is a low priority:

- Engagements are close range (atmospheric effects are more pronounced at long ranges)
- Heavy terrain effects (if there is a mountain in the way, the atmosphere probably won't bend your signal over the top of it)
- Close urban environments or dense foliage (you have too many confounding factors, and it's not worth adding more variables)
- Forecast shows benign conditions for your AO (if the atmosphere is nominal, there's no point in downloading/using the data)

If you don't import any METOC, the model will assume a "standard atmosphere", which will predict "standard refraction" in the picture above.

6 Exporting refractivity from Builder

6.1 Import Weather Data

You need DOD PKI certs for this. If you aren't familiar with the following software, your Navy AG or METOC Marine can help. To get a formatted refractivity file, you'll need NRL Scenario **Builder v4.3 or higher.**

NIPR https://builder.nrl.navy.mil

SIPR https://builder.nrl.navy.smil.mil

6.1.1 Builder v4.7 or higher

Use the built-in CAGIPS METOC downloader in Builder using the Builder Tutorial as a guide.

6.1.2 Builder v4.3 – 4.6

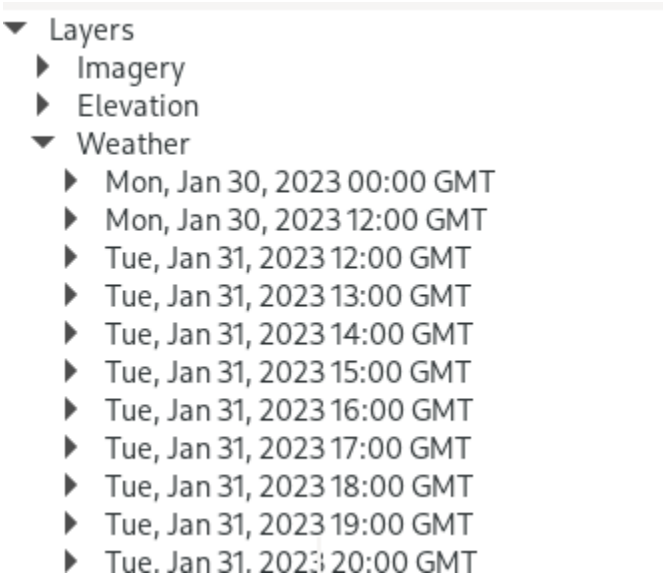
Download the GRIB2 file for your AOR:

NIPR https://portal.fnmoc.navy.mil - click COAMPS OS

SIPR https://portal.fnmoc.navy.smil.mil

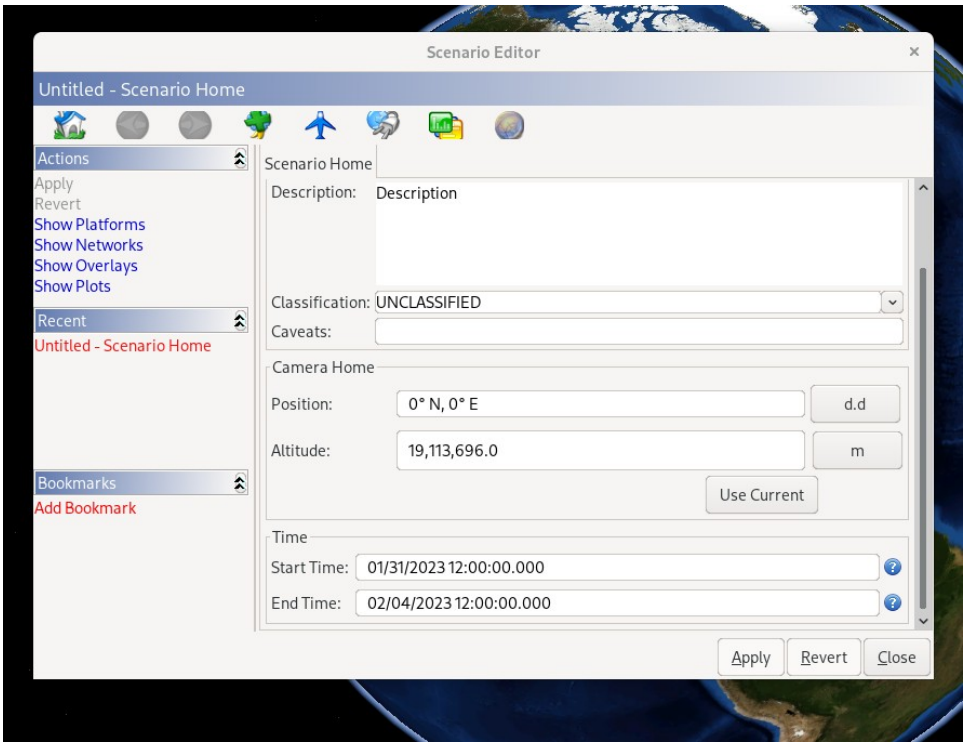
Once you start Builder, drag the GRIB2 file into the map window.

Verify the weather data has imported into Builder by selecting and expanding Layers – Weather. The METOC data should auto populate into the Weather layer.

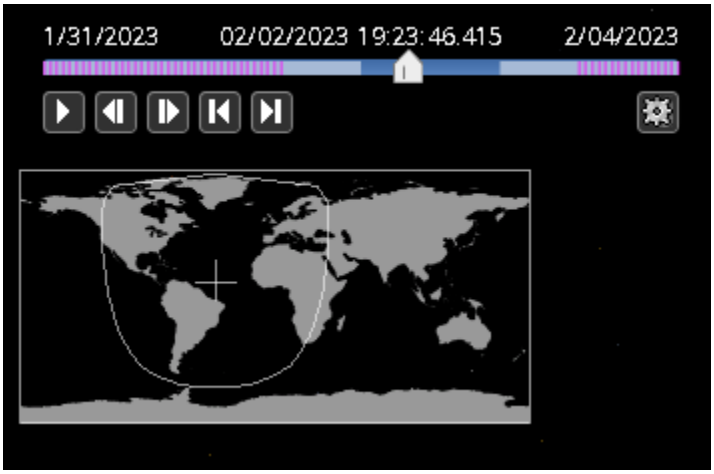


6.2 Select the time range

Edit -> Scenario Editor. Scroll to the bottom and set the Start Time and End Time to the time period you'd like to export (UTC timezone)



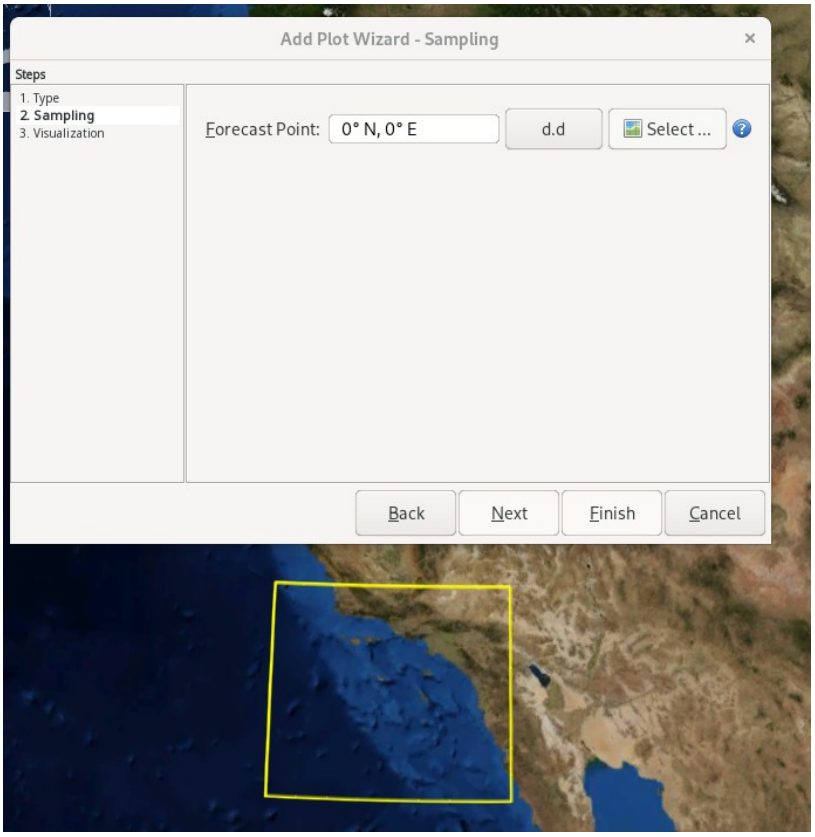
The purple hashmarks in the timeline show the regions where you have a valid forecast.



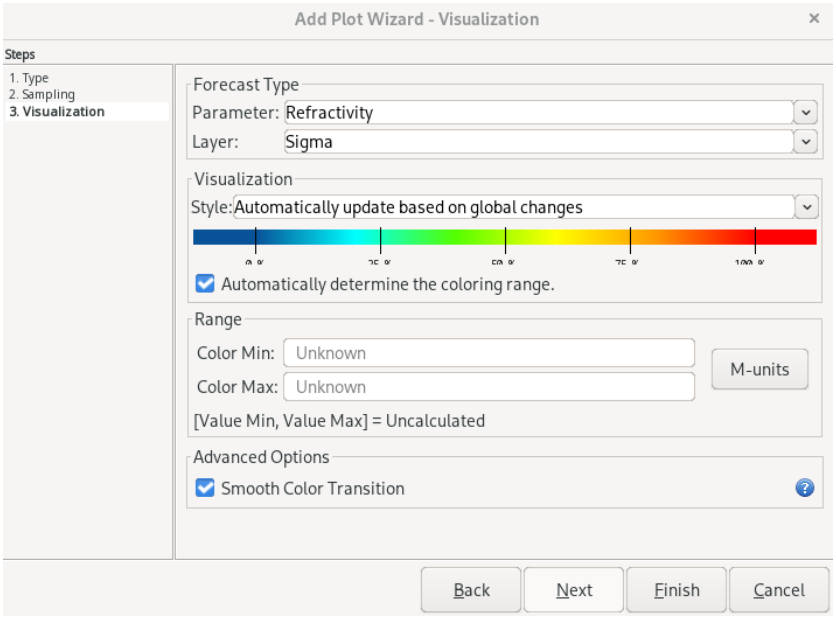
6.3 Create a refractivity plot

Edit -> Scenario Editor -> Show Plots -> Add Plot

Create a Forecast Plot and when choosing coordinates, use “Select” and click the map anywhere inside the bounds of your GRIB2 data:



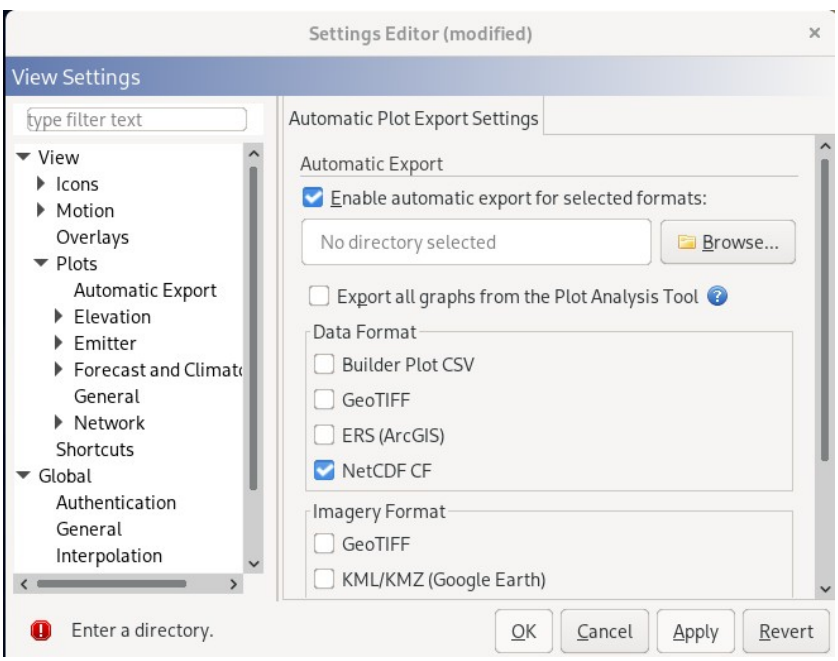
Choose Refractivity as the Parameter and click Finish.



6.4 Enable automatic plot exporting

Edit -> Settings Editor -> View -> Plots -> Automatic Export

Choose NetCDF CF as the Data Format, then choose a location to store the data.

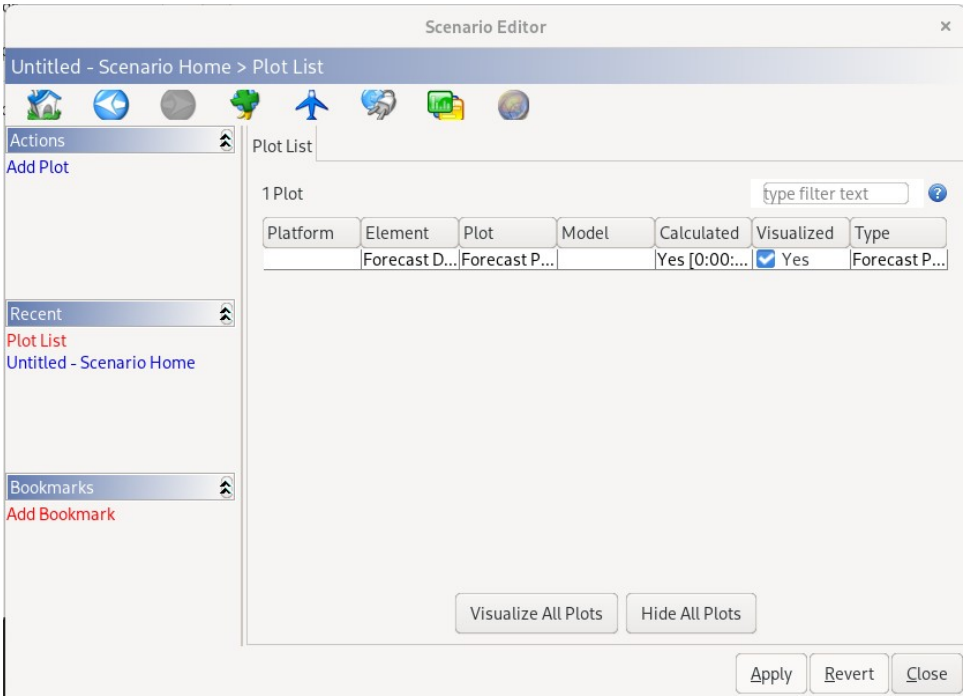


And click OK.

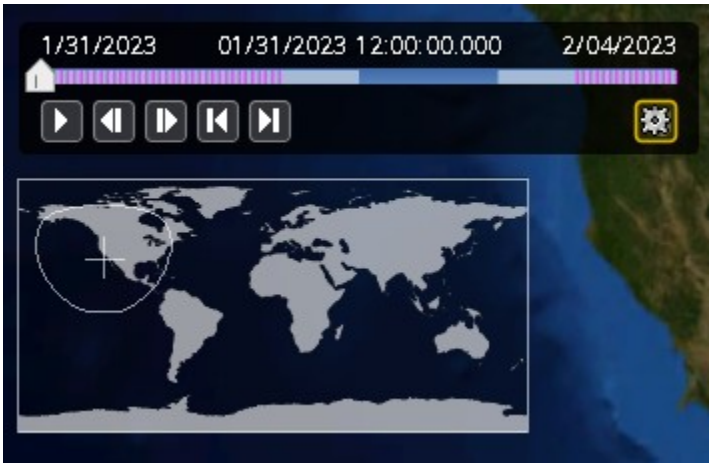
6.5 Enable plotting and play the scenario to export all forecast times

Go back to the Scenario Editor, and click Show Plots.

When you click on the Visualized checkbox, Builder will automatically export the refractivity data at the current time marker on the Timeline (should be the first one)



To automatically export the rest of the forecast times, we need to make the playback speed faster. Click the little gear underneath the Timeline, choose Playback Speed, and set to “1h”.



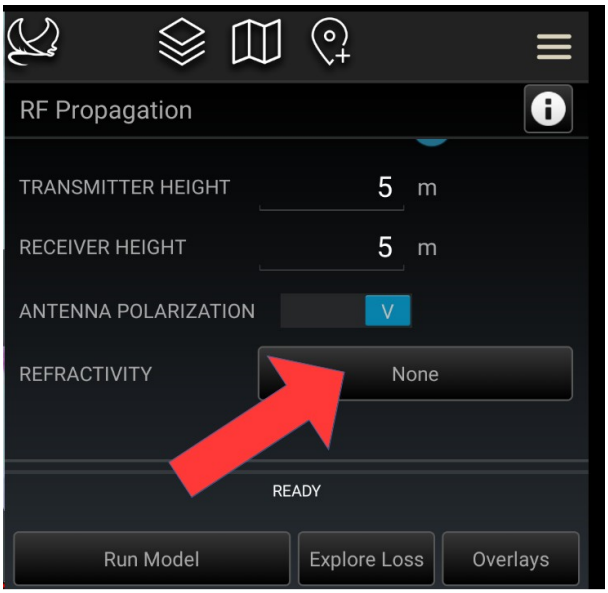
Now, press the Play button on the Timeline and the rest of the forecast times will export. This may take a while.

6.6 Push to ATAK device or TAK Server

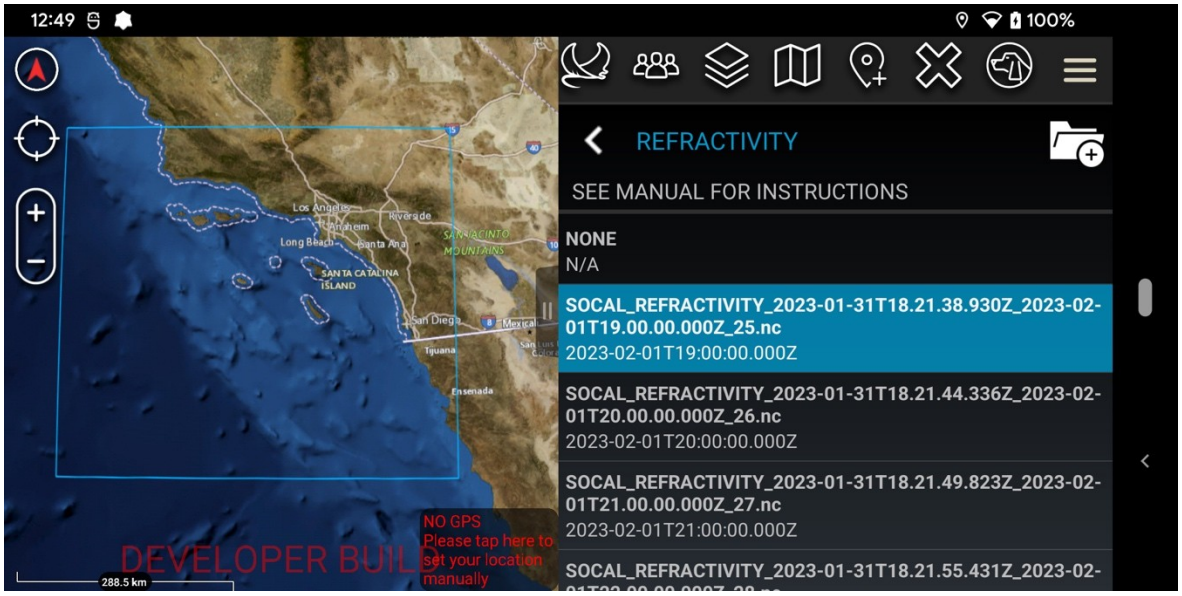
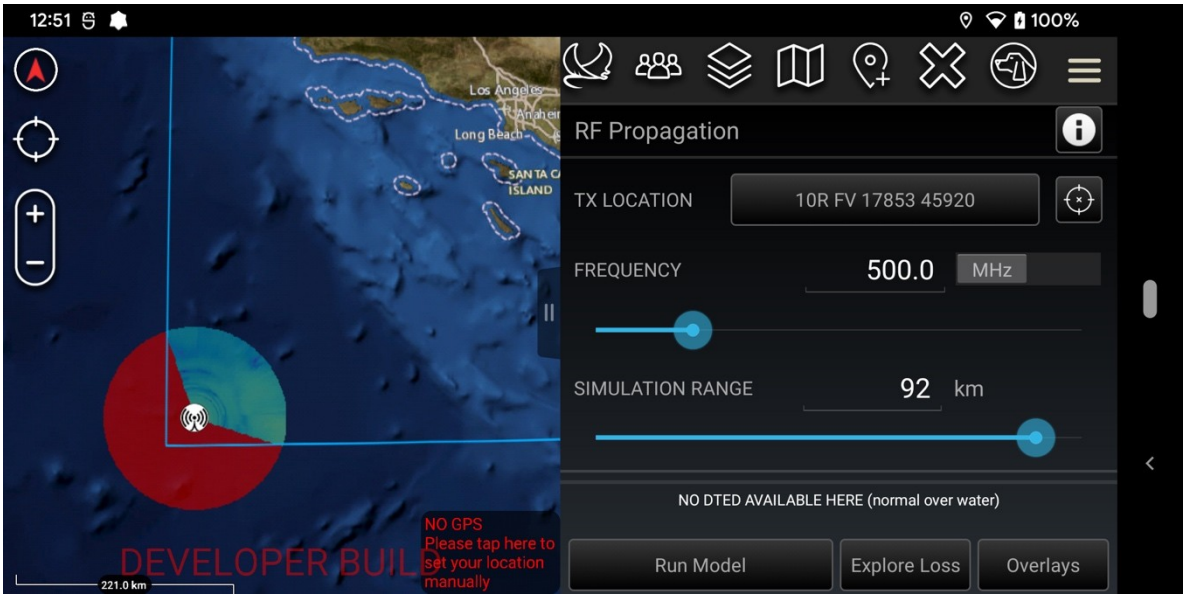
The set of netCDF files that were exported (.nc files) can now be sent to a device running ATAK.

7 Importing METOC Refractivity data in the plugin

In the RF Propagation Plugin, open the REFRACTIVITY view:



In this view, you can press the folder icon in the top right corner, and select the .nc files to be imported. Once they are imported, you can select the file to see a blue box around area that the forecast covers.



Any simulation run inside this box will use refractivity data for the selected forecast time. If you attempt to run a simulation outside the box, the data will be invalidated (dark reddish color):